

HOUSING PRICE PREDICTION

How different learning methods see the same data differently

Exploratory Data Analysis

Chapter 1 — First Look at the Scene

House value is capped at \$500,000—visible as a dense line at the top of every scatter plot.



Fig 1. Income vs. value. Clear upward trend, wide spread, cap visible at top.

Income is the strongest single predictor of value. House age shows almost no trend.



Fig 2. Age vs. value. No relationship.

Rooms, bedrooms, population, and occupancy all have extreme outliers (e.g. one averaging 140+ rooms/household).

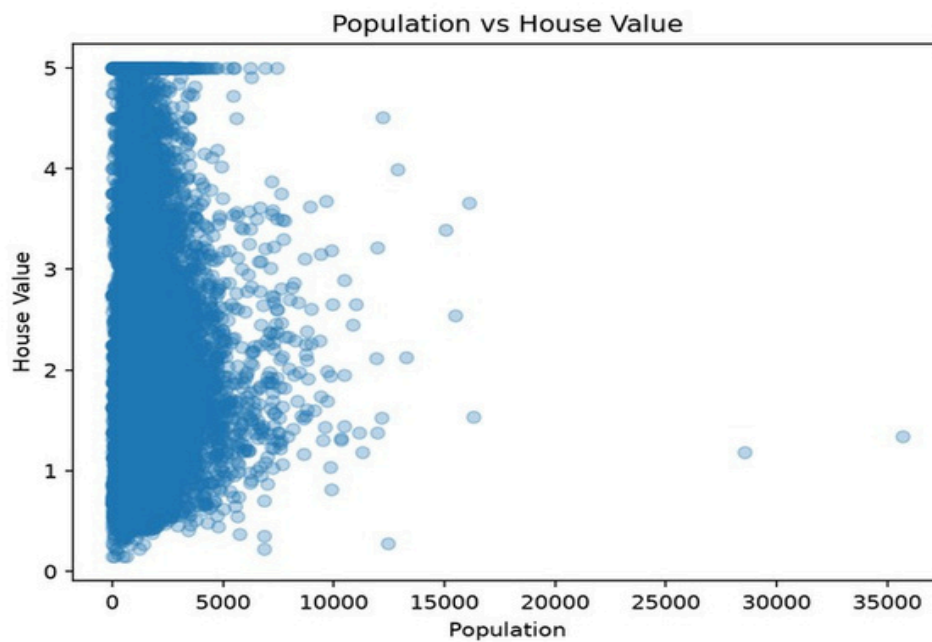


Fig 3. Population vs. value. Outliers stretch the axis.

Chapter 2 — Global Model vs. Local Model

Linear regression fits one flat surface across all the data. KNN adapts locally, with no fixed shape.

Linear Regression: MedInc + HouseAge

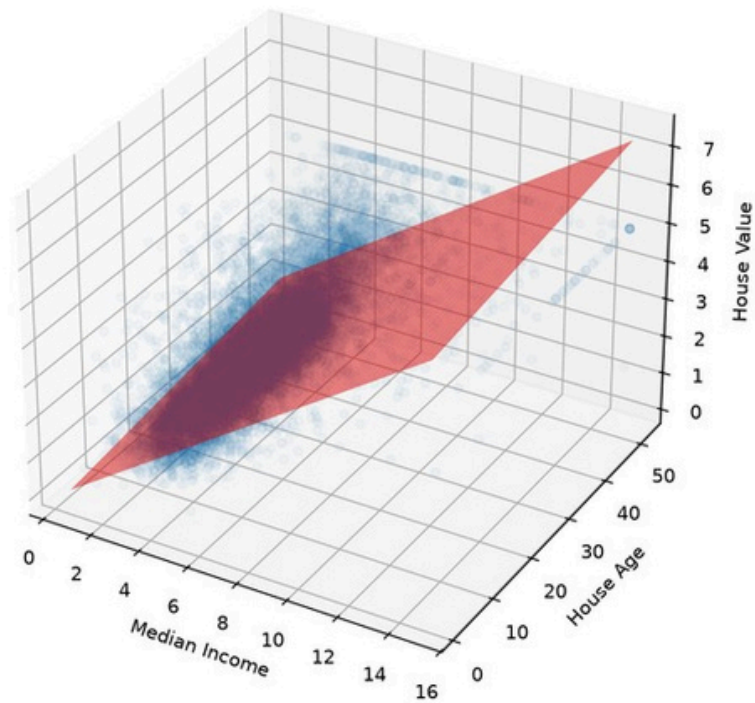
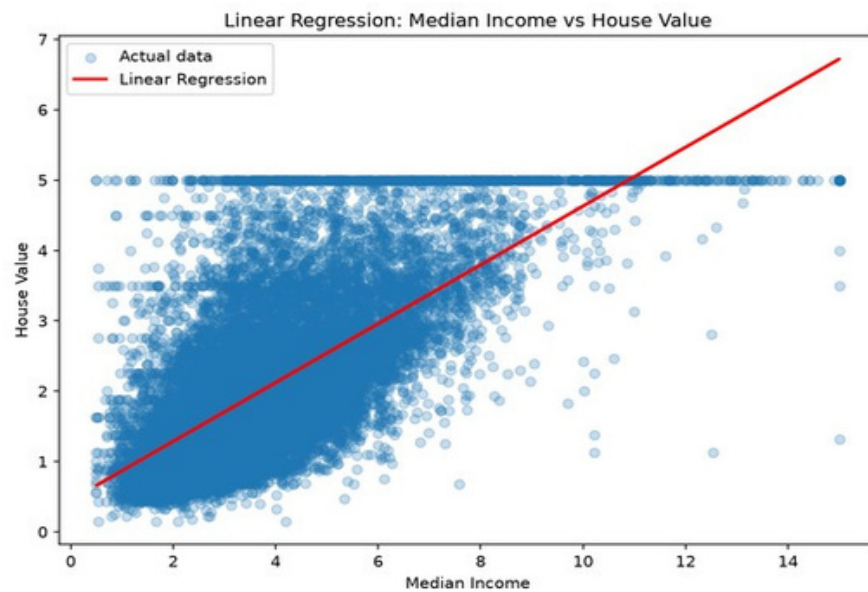


Fig 4. LR surface over income + age. One plane for the whole dataset.

On income alone: LR keeps rising past the price cap. KNN flattens out near the top, matching the real upper bound.



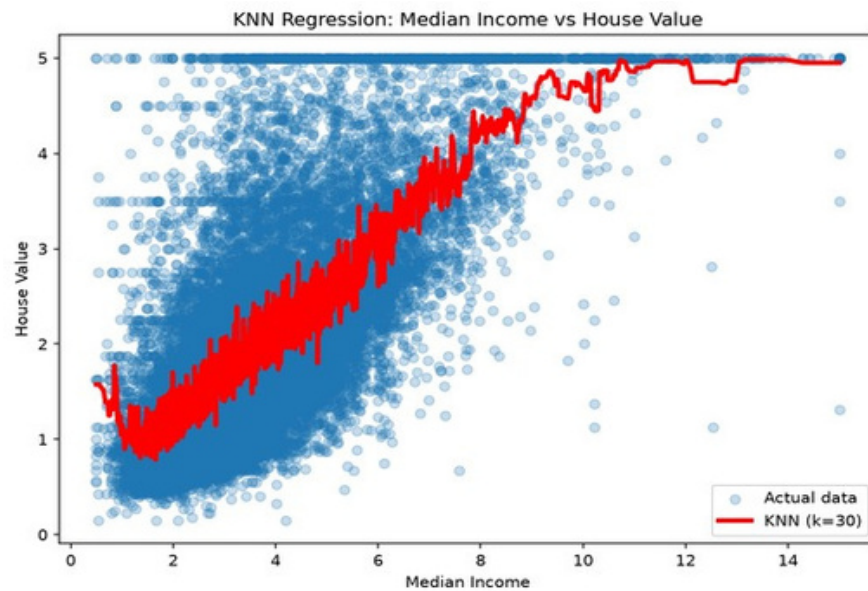


Fig 5 & 6. LR (top) vs. KNN k=30 (bottom). Agree in the dense middle, diverge at the extremes.

Chapter 3 — What Counts as Right?

Income subset: house values are skewed, mean > median.

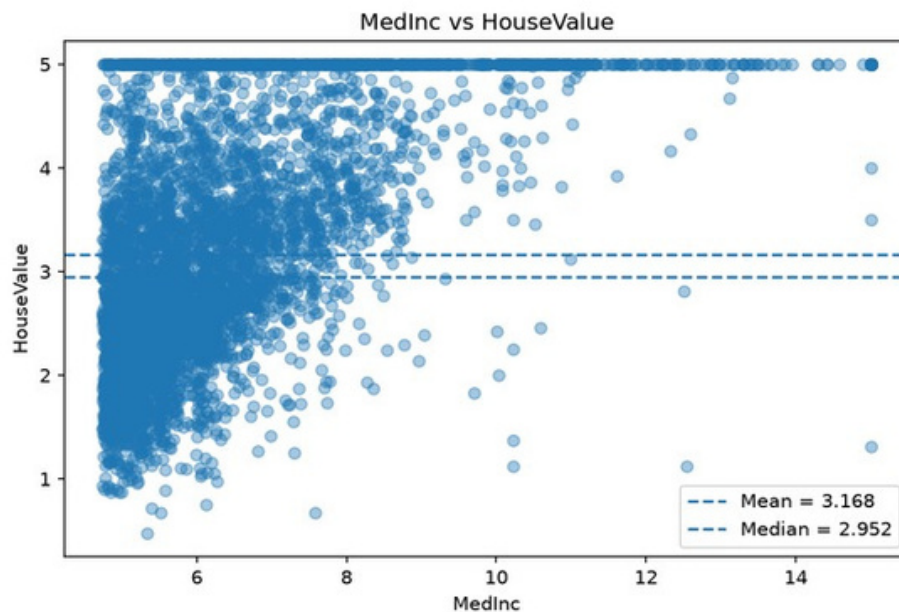


Fig 7. Wealthiest quarter of districts. Mean pulled above median by the price cap.

Squared loss is minimized at the mean; absolute loss at the median. "Best" prediction depends on which errors you penalize more.

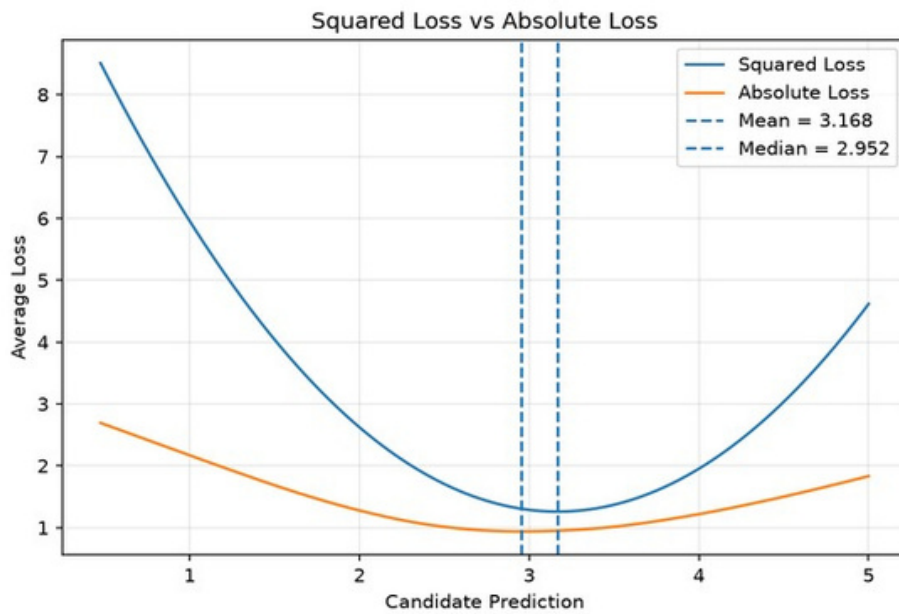


Fig 8. Squared loss (mean-seeking) vs. absolute loss (median-seeking).

Chapter 4 — When Neighbors Stop Being Local

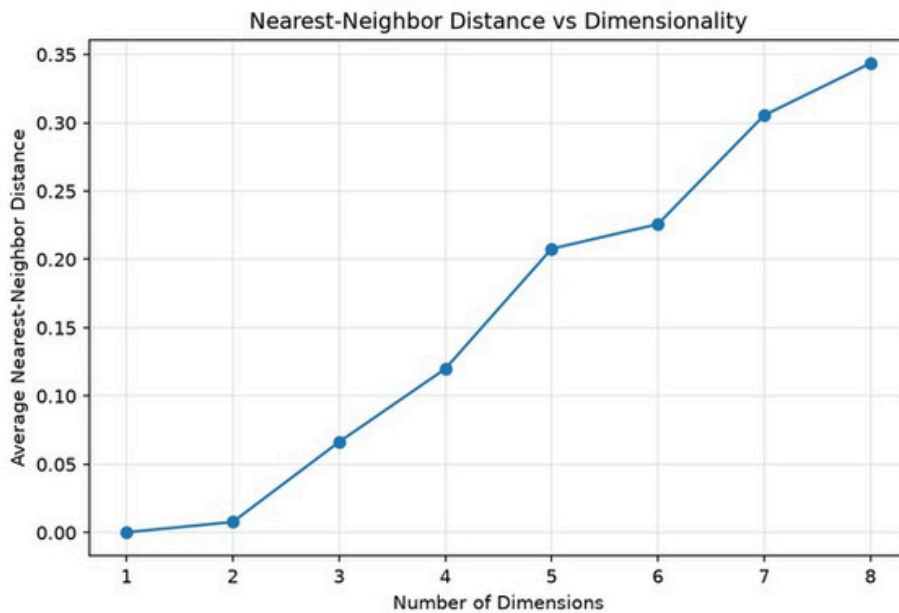


Fig 9. Nearest-neighbor distance grows steadily with added dimensions.

As expected: nearest-neighbor distance rises with dimension — the classic curse of dimensionality.

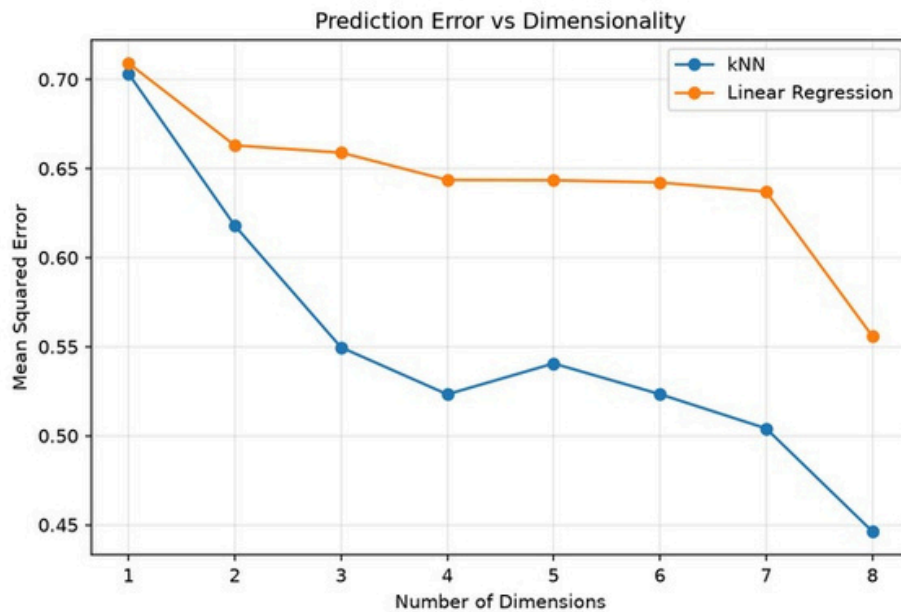


Fig 10. Test error falls for both models as dimensions increase; KNN pulls ahead.

Unexpected: error falls for both models, and KNN improves most, once latitude/longitude are added. These added dimensions carry real signal (location), not noise — so the usual curse doesn't apply here. The curse applies to noised and useless dimensions, not dimensions in general.

Chapter 4b — Adding Structure (Optional)

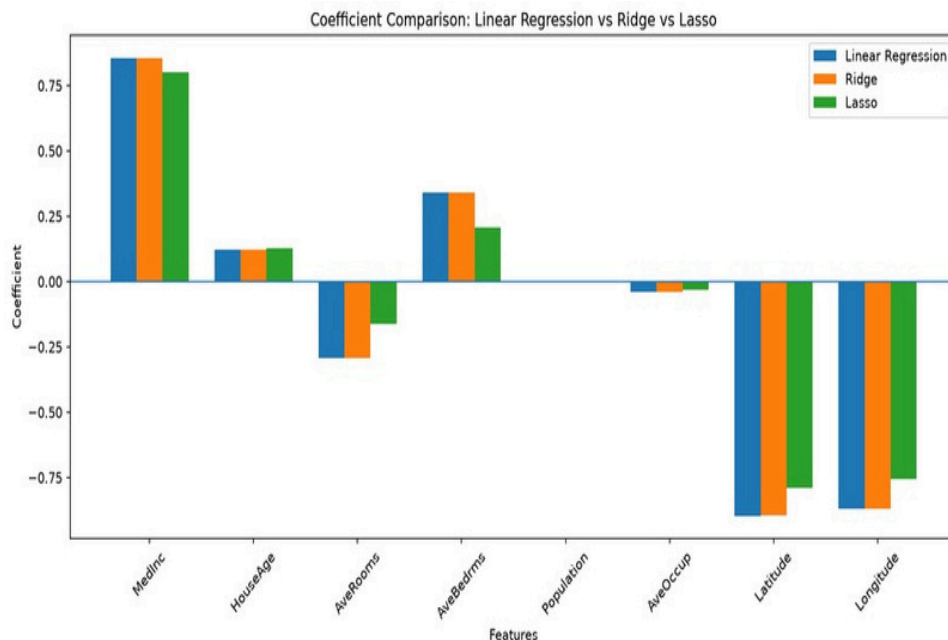


Fig 11. OLS vs. Ridge vs. Lasso coefficients. Income and location dominate; Lasso shrinks weak features hardest.

All three models agree on direction and on which features matter most. Lasso trims weak features (rooms, bedrooms) toward zero.

Chapter 5 - The story

As you can see the coefficients of lat, long are comparatively higher than others, so my final model takes only these features into consideration for prediction.

I have used these features and fitted LR and KNN(k=20) and got the following results or values for MSE and R^2 :

LINEAR REGRESSION :

MSE : 0.97884357877149

R^2 : 0.25302436489293967

KNN(k=10):

MSE : 0.270726741943905

R^2 : 0.7934028639613488

Therefore KNN with k=10 is performing better and is thus my final model.

As I have proceeded throughout the tasks, I have understood which features are important using plots and intuition, understood which model, loss, and features to be taken so that we can get maximum accuracy.

Like initially I thought medianInc is important thru the graph as other graphs were pretty scattered or too linear, then saw that increasing dimensions in KNN increases distance and makes all points more or less similar far apart, also seen that error reduces more at latitude and longitude including part in dimensions and when took them first the error turned out to be pretty constant making it clear that latitude and longitude are pretty important as well.

Also I took coefficients of all features and saw that lat and lon have pretty significant coeff but others didn't, so for my final model I picked lat and long as features and performed both LR and KNN and got better results for KNN.